Tobruk University / Faculty of Science

Midterm Examination 2025-2026 **Department of Physics**

Module: Vibration and waves. **Course Code:**

Time: 2:00 Hours **Max. Marks: 70**

Date: 4 / 2/2026 **Prepared by: Mohamed Y Bakkar**

Page numbers: 2

Q1- Define the following:

1. Oscillation 2. Momentum 3. Frequency 4. Simple harmonic motion (SHM)

.....(12 marks)

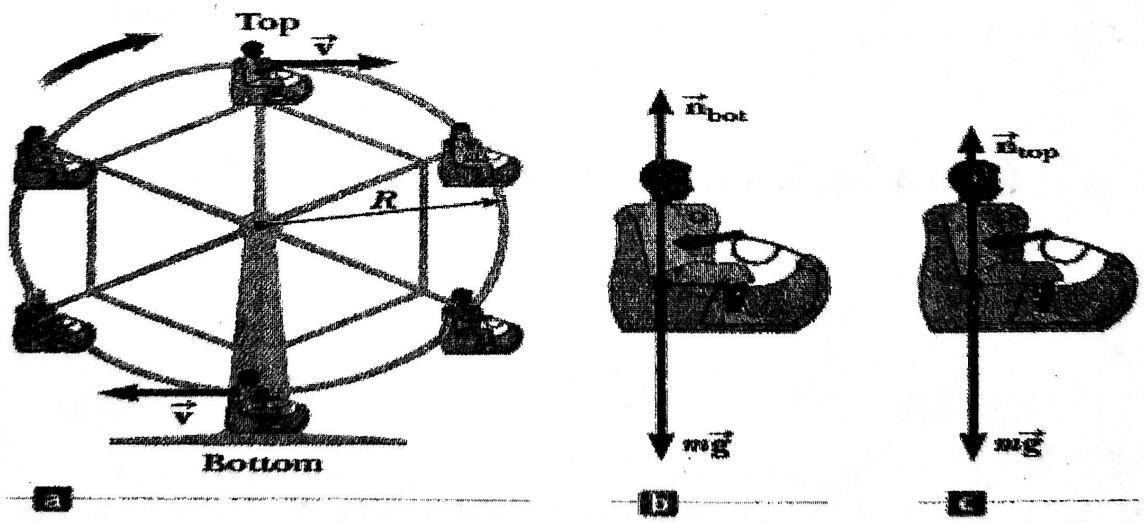
Q2- What are the properties of oscillation?

.....(15 marks)

Q3- A child of mass m rides on a Ferris wheel as shown in figure below, the child moves in vertical circle of radius 10m at constant speed of 3m/s.

A) Determine the force exerted by the seat on the child at the bottom of the ride, express your answer in terms of the weight of the child mg.

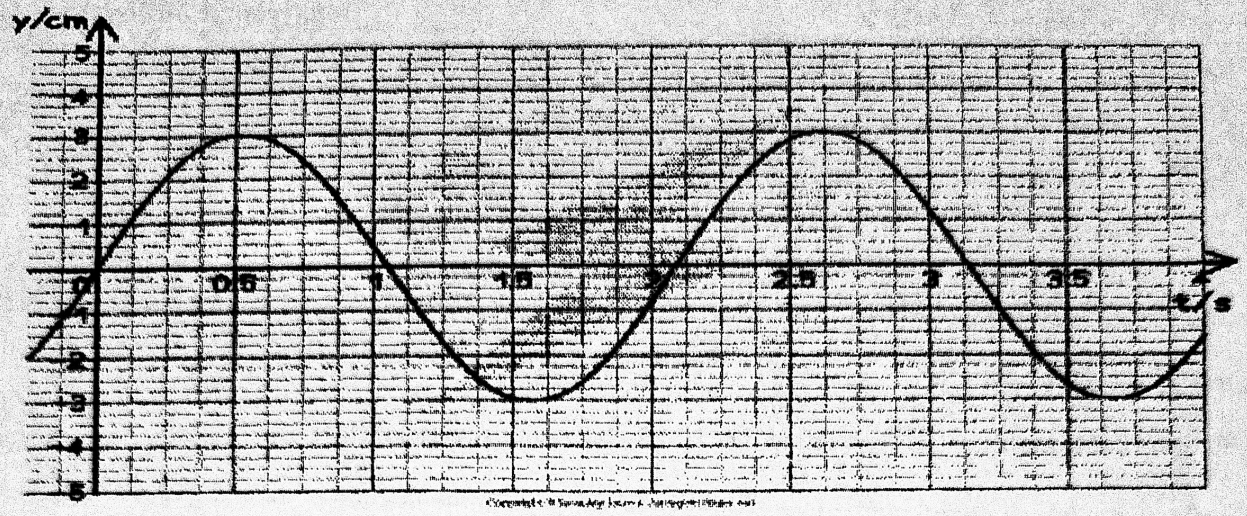
B) Determine the force exerted by the seat on child at the top of the ride.



.....(15 marks)

Q4-

A. A student sets out to investigate the oscillation of a mass suspended from the free end of a spring. The mass is pulled downwards and then released. The variation with time t of the displacement y of the mass is shown in the figure below.



B. State that some of examples of oscillators that undergo SHM (simple harmonic motion)?

.....(15 marks)

Q4- A mass of 55 g is suspended from a fixed point by means of a spring. The stationary mass is pulled vertically downwards through a distance of 4.3 cm and then released at t = 0.

The mass is observed to perform simple harmonic motion with a period of 0.8 s. Calculate the displacement x in cm of the mass at time t = 0.3 s.

.....(13 marks)

Good luck for exam & do the best